Noise-Augmented Urban Sound Classification under Matched and Mismatched Conditions

ELEC5305 Speech and Audio Processing — Research Report

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Abstract

Environmental sound classifiers are routinely exposed to interfering sounds that differ from those available during model development. This project asks: To what extent does waveform-level noise augmentation improve an MFCC–SVM urban sound classifier when the added test noise is matched to, or different from, that used during training? A reproducible ten-fold evaluation was conducted on all 8,732 UrbanSound8K clips using a 225-dimensional MFCC-based representation and an RBF support vector machine. Four training strategies were compared: original audio, white-noise augmentation, traffic-noise augmentation, and a mixture of white, pink, and traffic noise. Each model was tested on original recordings and on four added-noise conditions at 20, 10, and 0 dB signal-to-noise ratio (SNR). The real traffic and held-out crowd environments were drawn from DEMAND recordings. Across the 12 added-noise conditions, mixed augmentation increased mean fold-level accuracy from 50.23% to 59.91% and macro F1 from 47.66% to 61.90%. However, traffic augmentation outperformed mixed augmentation in the held-out crowd environment at all three SNRs, including 42.71% versus 39.59% accuracy at 0 dB. Detailed errors reveal pronounced class imbalance in predictions under severe noise. A supplementary analysis of four selected clips (48 perturbations) shows that relative MFCC distortion increases as nominal SNR falls; at 0 dB, mean distances were 0.662 for white, 0.542 for pink, 0.410 for crowd, and 0.330 for traffic noise. These four-clip observations are not population-level explanations of classification outcomes. The findings support noise augmentation as a useful, but distinctly noise-dependent, robustness intervention; they do not support a claim of uniformly improved generalisation to unseen acoustic environments.

Keywords— environmental sound classification; UrbanSound8K; data augmentation; MFCC; support vector machine; additive noise; domain mismatch; robustness.

1. Introduction

Urban acoustic recordings contain overlapping sound sources, background activity, differences in microphone placement, and variations in recording quality. An automatic classifier trained on one acoustic distribution may therefore degrade when deployed in another. This is particularly problematic for practical urban-sound recognition, where an engine, siren, horn, and human activity can occur simultaneously. UrbanSound8K provides a common, annotated ten-class dataset and official recording-aware cross-validation folds for studying such challenges [1].

A common solution is to augment training recordings with acoustically plausible disturbances, thereby teaching a classifier to tolerate variation that it may encounter at test time. In deep learning, audio augmentation is often associated with better generalisation, although benefits differ by augmentation type and sound class [4]. Here, instead of replacing the classifier with a high-capacity neural network, the feature extractor and classifier are held constant: a reproducible MFCC representation coupled to an RBF-SVM. The controlled variable is the training-noise regime. This design allows a focused investigation of robustness gained from different noise distributions rather than primarily measuring the benefit of architectural complexity.

The **research question** is whether waveform-level noise augmentation improves classification in *matched*, *mismatched*, and *previously unseen* noise conditions, and how this benefit changes with SNR. Three working expectations were assessed. First, training with a particular noise type should benefit testing in that same noise. Second, mixing different training noises might improve average performance across a broad test suite. Third, greater training-noise diversity might improve performance on a held-out real recording; importantly, this third expectation is tested rather than assumed.

This report presents the prior work, methods, ten-fold results, class-level failure analysis, and methodological limitations. It distinguishes **original recordings** from acoustically *clean* audio: UrbanSound8K contains real field recordings and may already include unwanted sound sources. Thus, a nominal 0 dB SNR here specifies added-noise power relative to the original waveform, not the ratio between the labeled foreground sound and all ambient noise.

2. Literature Review

2.1 Dataset design and classical MFCC classifiers

Salamon, Jacoby, and Bello introduced the UrbanSound taxonomy and UrbanSound8K subset to address a shortage of shared, realistic urban-sound annotations [1]. Their experiments used MFCC descriptors and several classifiers, including an RBF-SVM, with considerable attention to the correlation between multiple audio slices from the same source recording. This motivates the present use of the **official predefined folds** rather than an independently shuffled clip-level split. Ignoring recording families

would risk leakage and optimistic accuracy estimates. Their work also highlighted confusions between acoustically similar categories, such as air conditioning and engine idling, and sensitivity to background interference.

MFCCs compress a short-time spectral representation into coefficients associated with perceptual frequency scaling. Davis and Mermelstein established the value of mel-frequency cepstral features for speech recognition [2], while later environmental sound systems adapted cepstral summaries to non-speech acoustic events [1,7]. The appeal is a relatively compact, interpretable, low-cost representation. However, because MFCCs are derived from short-time spectral energy, additive noise changes log-band energies and may shift feature vectors in ways that a classifier trained on quieter or different backgrounds has not encountered. Additionally, clip-level summary statistics compress potentially important temporal structure.

The original UrbanSound8K paper describes 25 MFCC coefficients computed from 40 mel bands using an approximately 23.2 ms analysis window, 50% overlap, and a 225-dimensional summary [1]. Its stated list of statistics admits a small ambiguity when counted literally. The primary implementation here uses 7 per-coefficient statistics (minimum, maximum, median, mean, variance, skewness, kurtosis) and the means of the first and second derivatives, yielding **25 × 9 = 225** features. Since this project uses *librosa* and *scikit-learn* rather than the original Essentia and Weka implementations, its baseline is a **methodologically close approximation**, not a numerically exact replication of the published model. Absolute accuracy should not be compared as though the software pipelines were identical.

2.2 Robust representations and mismatched noise

Dennis, Tran, and Li proposed a spectrogram-image feature specifically for sound-event classification under mismatched conditions [3]. Their motivation is that time-frequency structures associated with a target sound may persist despite diffuse interference, whereas some conventional descriptors change substantially when noise is introduced. Their work provides a useful contrast to the present project. One route toward robustness changes the *representation* itself; the present study instead keeps the representation fixed and changes the *training distribution*. This difference matters when interpreting both success and persistent failures: an SVM cannot recover acoustic information that has already been strongly obscured or collapsed into unsuitable MFCC statistics.

2.3 Data augmentation and model complexity

Salamon and Bello demonstrated the use of data augmentation together with convolutional neural networks for environmental sound classification [4]. Their study is not a direct performance benchmark for the present system: its representation, capacity, training procedure, and augmentation mechanisms differ. It nevertheless supports two important principles: the augmentation mechanism should be evaluated empirically, and its impact may be class-dependent. Piczak's spectrogram-based convolutional model also illustrates the alternative strategy of learning time-frequency patterns with a more expressive architecture [6]. These studies motivate a bounded project scope: rather than claiming state-of-the-art classification, the current work isolates a specific question about augmentation transfer in a simpler model.

2.4 Real recorded noises and research gap

The Diverse Environments Multichannel Acoustic Noise Database (DEMAND) contains real background recordings from contrasting scenes [5]. The selected STRAFFIC traffic recording provides an environmental training perturbation; PCAFETER (crowd/cafeteria) is deliberately excluded from augmentation training and serves as one held-out noise environment. This supports a test that is more realistic than using only synthetic white noise, while also limiting the conclusions: one held-out scene and one selected channel are not a diverse population of unseen real environments.

Together, the literature establishes that (i) recording-aware evaluation is essential, (ii) additive interference can distort fixed cepstral representations, and (iii) augmentation may or may not transfer between acoustic conditions. The **specific contribution** of this project is a reproducible, equal-protocol comparison of original, single-noise, and mixed-noise MFCC–SVM training across matched, mismatched, and held-out tests, supplemented by a fold-level and class-level examination of where transfer fails.

3. Methodology

3.1 Data and preprocessing

The primary dataset is UrbanSound8K, containing **8,732** labeled audio clips from ten urban sound categories [1]. All ten official folds were evaluated. For each test fold, the remaining nine folds formed the training set, and the complete model preprocessing sequence was fitted using training-fold data only. The original recordings were loaded and resampled to **44.1 kHz mono** for feature extraction. Two DEMAND environmental recordings were extracted from the 48 kHz archives by selecting the ch01.wav channel and then resampled by the project audio loader [5]. The selected file identities and deterministic seed (540334953) are preserved by the experiment configuration.

For an environmental recording, its initial **60%** was reserved for training-noise excerpts, a **10%** intervening temporal guard separated regions, and the final **30%** was reserved for test excerpts. The PCAFETER environment was never used during training. This reduces same-recording temporal overlap but does not imply independence of every acoustic event across time or generalisation to multiple environments.

3.2 Controlled waveform noise mixing

Let the source waveform be $s[n]$ and a selected zero-mean noise excerpt be $v[n]$. The implementation scales the noise according to full-clip mean-square powers P_s and P_v:

$$\alpha = \sqrt{\frac{P_s}{P_v 10^{SNR_{dB}/10}}}, \; y[n] = s[n] + \alpha\, v[n].$$

Mixing occurs in floating point. If necessary, a single common gain is applied to the entire mixture to keep its amplitude within the output range; the code does not apply nonlinear clipping or independent source normalisation. Test perturbations were created at **20, 10, and 0 dB**. White and pink noise were synthesized; traffic and crowd excerpts came from DEMAND. The random selection of noise segments and levels was seeded using the clip identity and experimental condition, ensuring reproducible conditions across models.

3.3 Audio representation and classifier

Features were calculated with 1,024-sample frames, 512-sample hops, 40 mel filter-bank channels, and 25 MFCC coefficients. Seven temporal statistics per coefficient and the mean first and second derivative coefficients produce a **225-dimensional clip vector**. An RBF-SVM was trained with C=1.0 and gamma=scale. Training-only fitted scaling and feature-selection steps were included in the model pipeline, avoiding test-to-train information leakage. The feature-extraction configuration remained identical across all training strategies [1–3].

Four training policies were compared. **Original** uses unmodified training clips. **White** retains original clips and adds one copy per clip mixed with white noise. **Traffic** does the same with STRAFFIC. **Mixed** retains original clips and adds one copy per clip, choosing one of white, pink, or STRAFFIC noise. In each augmented policy, the SNR of that single added copy is a seeded selection from 0, 10, and 20 dB. A key design limitation is that all three augmented policies effectively double the number of training instances, whereas Original does not. Thus, an augmented-versus-original performance difference is not a pure causal estimate of *noise type alone*.

3.4 Testing and evaluation

Each trained strategy was evaluated on unmodified clips and on four added-noise conditions (white, pink, traffic, crowd) at three SNRs, for **13 conditions per strategy per fold**. There are **4 × 10 × 13 = 520** fold-level records. For each condition we report fold-averaged accuracy, macro F1, and macro recall. Macro F1 is relevant because accuracy alone can obscure severe deterioration in under-recognised classes. Aggregate noisy-test summaries give equal weight to the **12 noise/SNR combinations and 10 official folds**, rather than pooling all predictions across unequal fold sizes.

All 520 fold-level metric records were checked against 454,064 out-of-fold predictions, with agreement at numerical precision. No duplicate condition/fold records or invalid metric values were found in the supplied output. The original/unmodified condition in the full experiment agreed with the independently completed baseline run. Paired, exploratory 95% t intervals were computed from ten fold-level differences, averaging repeated noise conditions within each fold before inference; the heavily overlapping training portions mean these are descriptive intervals rather than strict independent-sample population claims. A separate, descriptive signal-analysis check selected one Fold 1 clip from each of four classes and analyzed all four noise types at 20, 10 and 0 dB (48 mixtures). Its per-condition waveform, log-Mel and MFCC plots, achieved SNRs, and relative representation distances were saved alongside the classification metrics; no classifier was retrained for this supplement.

4. Results

4.1 Baseline and original-recording performance

The independent, unaugmented ten-fold baseline reached **64.41% accuracy** (fold standard deviation 5.01 percentage points), **65.63% macro F1** (4.23 points), and **66.37% macro recall** (4.22 points). The performance of each augmented model on the *original* recordings remained very similar (Table 1). Thus, the robustness benefits discussed below were not accompanied by a large average penalty on the original test condition; however, such small differences are modest relative to fold-to-fold variability.

Table 1. Original-recording (no newly added noise) performance, ten-fold mean.

Training strategy	Accuracy (%)	Macro F1 (%)
Original	64.41	65.63
White	64.12	65.15
Traffic	64.97	65.90
Mixed	64.35	65.34

4.2 Overall robustness on added-noise tests

Across all four noise types and three test SNRs, mixed augmentation achieved the highest mean performance (Table 2). Its mean accuracy increased from **50.23% to 59.91%** compared with Original, a gain of **9.68 percentage points**, while macro F1 increased by **14.24 points**. The white-only and traffic-only strategies also improved mean accuracy, but the pooled result conceals important variations by noise source and SNR.

Table 2. Equal-weight average across the 12 added-noise test settings and ten folds.

Training strategy	Accuracy (%)	Macro F1 (%)	Accuracy gain vs Original (pp)
Original	50.23	47.66	—
White	58.54	60.23	+8.31
Traffic	56.08	57.36	+5.85
Mixed	**59.91**	**61.90**	**+9.68**

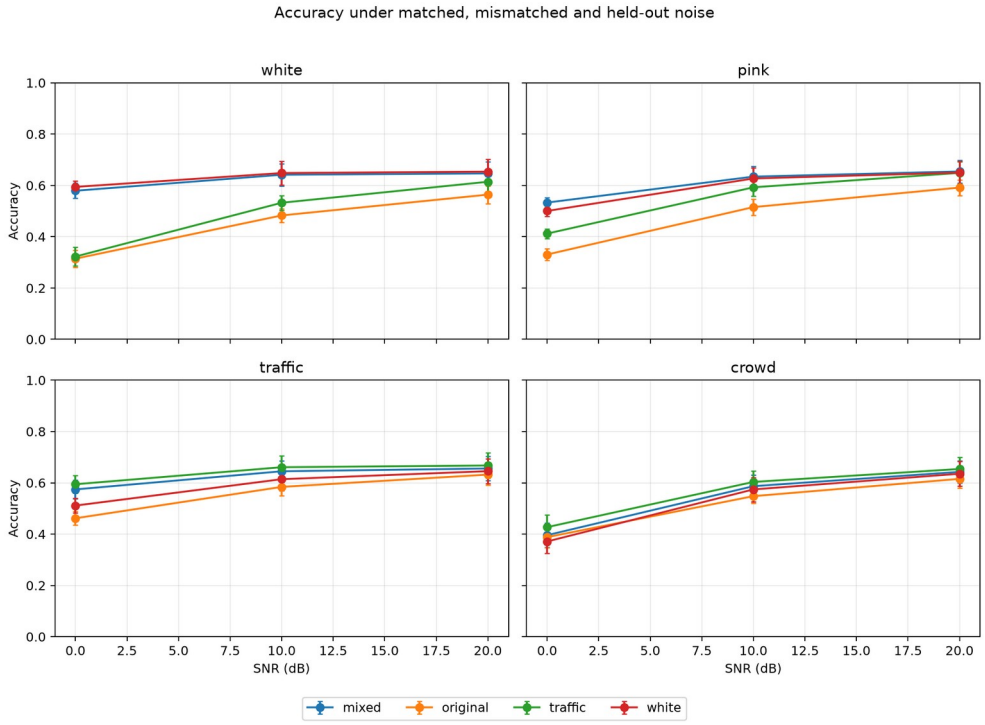


Figure 1. Accuracy versus test SNR for the four training strategies under white, pink, traffic, and held-out crowd noise. Error bars represent the standard deviation across ten folds, as produced by the project plotting code.

At 0 dB averaged across the four noise types, Mixed reached **52.06%**, compared with **37.37%** for Original, **49.43%** for White, and **43.91%** for Traffic. The positive Mixed–Original difference on the full noisy test suite occurred in **all ten folds**, with a mean +9.68 pp and an exploratory fold-paired 95% interval of **[+7.76, +11.60] pp**. These observations support a robust average improvement *within the evaluated settings*.

4.3 Matched and mismatched noise

The 0 dB transfer matrix exposes the role of noise matching (Table 3 and Figure 2). White-only training performed best on 0 dB white noise (**59.42%**), while traffic-only training performed best on 0 dB traffic noise (**59.44%**). Mixed was strongest on 0 dB pink noise (**53.31%**), even though this was only one component of its randomized mixed augmentation. Thus, the broadest training regime provides a useful compromise across synthetic test noises but is not universally optimal.

Table 3. Accuracy (%) for the hardest added-noise conditions (0 dB); ten-fold average.

Test noise	Original	White	Traffic	Mixed
White	31.45	**59.42**	32.26	57.92
Pink	33.06	50.04	41.22	**53.31**
Traffic	46.17	51.12	**59.44**	57.42
Crowd (held out)	38.82	37.13	**42.71**	39.59

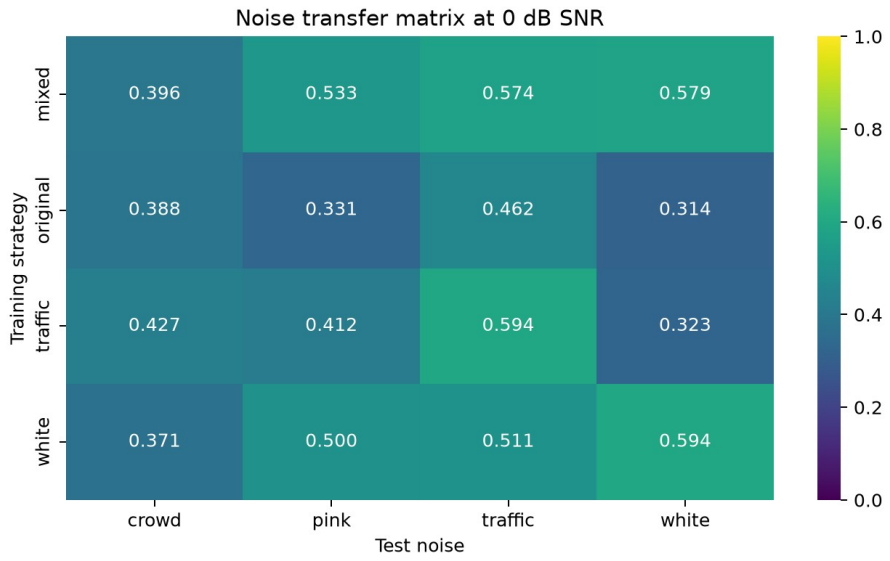


Figure 2. Observed accuracy transfer matrix at 0 dB SNR. Rows specify training policies; columns specify added test noise. The DEMAND PCAFETER crowd recording was held out during augmentation training.

4.4 Held-out environment and class-level failure modes

The strongest challenge to a simple "more noise diversity is always better" explanation appears in the single held-out crowd environment. Traffic augmentation exceeded Mixed at **20 dB (65.40% versus 64.25%)**, **10 dB (60.34% versus 58.68%)**, and **0 dB (42.71% versus 39.59%)**. The average Traffic–Mixed difference across the three crowd SNRs was **+1.98 pp** with an exploratory fold-paired interval of **[+1.11, +2.84] pp**. The 0 dB difference was **+3.13 pp**, positive in all ten folds. This is a consistent result for *PCAFETER*, not evidence that traffic training necessarily generalises to every novel environment.

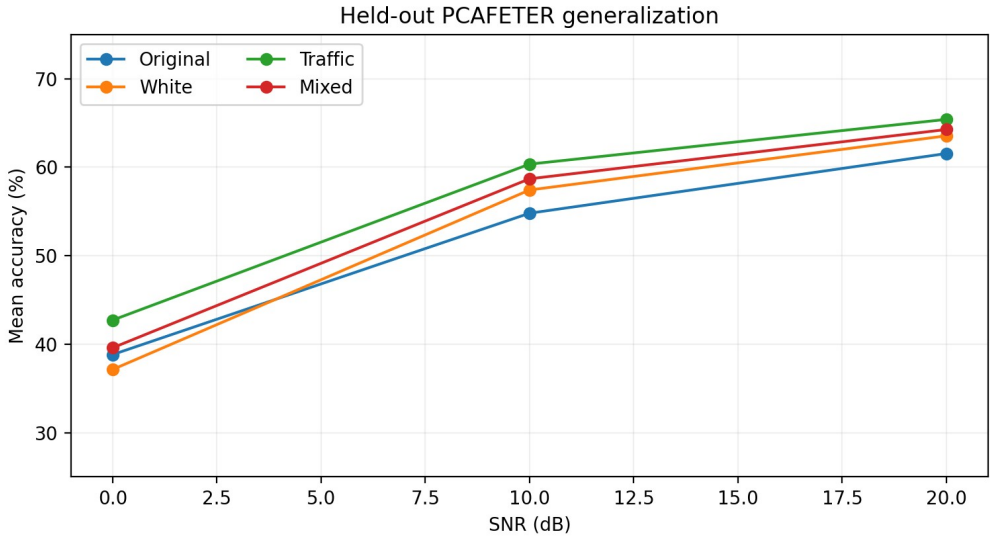


Figure 3. Held-out PCAFETER accuracy by SNR, recomputed from the supplied fold-level records.

Class-level predictions reveal an additional weakness. In the held-out crowd recording at 0 dB, the Mixed model assigned **48.92%** of clips to street_music, although that class accounted for only approximately **11.45%** of test examples. Its recall for street_music was **91.6%**, yet recall was only **6.1%** for engine_idling, **16.0%** for air_conditioner, and **24.0%** for drilling. Severe interference therefore produced strongly asymmetric errors that cannot be understood from overall accuracy alone. The traffic model had **50.0%** recall on gun_shot in this setting, compared with **43.6%** for Mixed and approximately **0.5%** for Original.

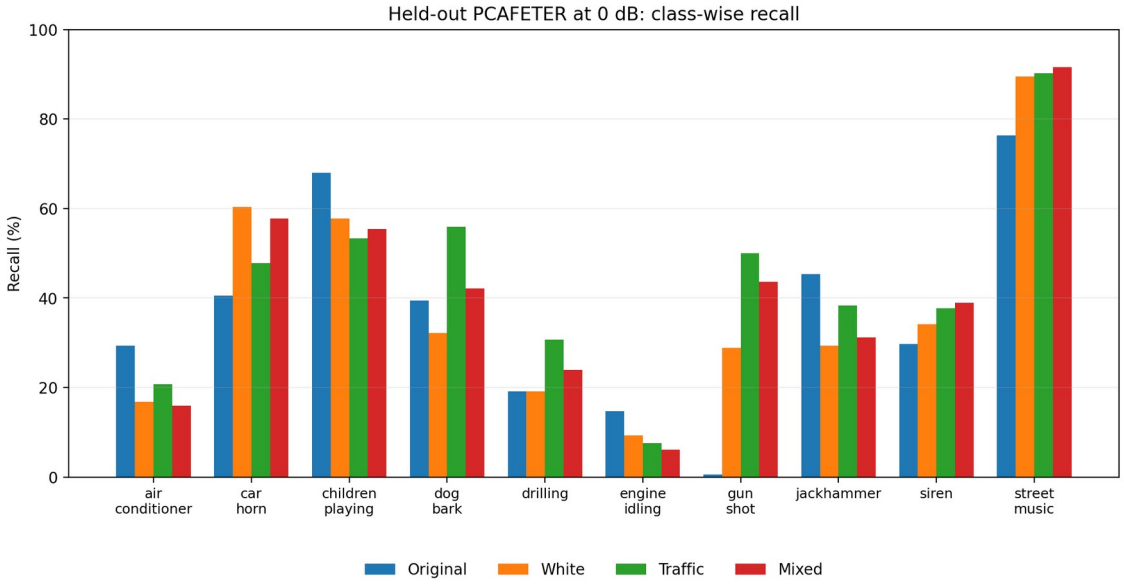


Figure 4. Ten-class recall in the held-out crowd environment at 0 dB, calculated directly from individual predictions (illustrative comparison, not a separate training experiment).

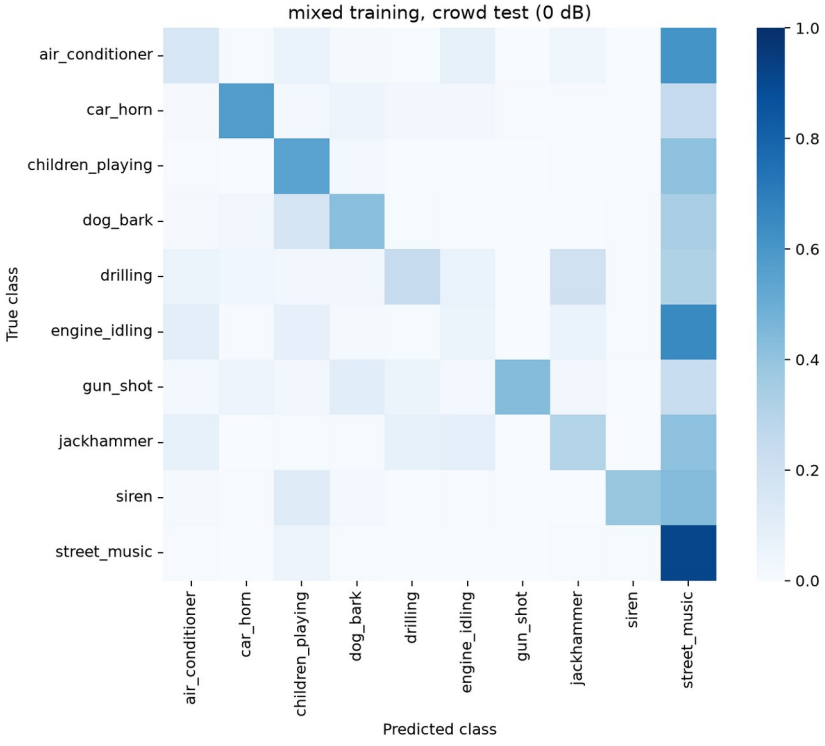


Figure 5. Row-normalized confusion matrix for the Mixed model under held-out crowd noise at 0 dB. The visually strong street_music prediction column is consistent with the class-level counts.

4.5 Acoustic representation analysis on four selected clips

To supplement the classifier-level outcomes with signal-processing evidence, the existing audio-analysis program was run on one deterministically selected Fold 1 clip from each of four classes: siren (106905-8-0-0.wav), street_music (108041-9-0-11.wav), engine_idling (103258-5-0-0.wav), and gun_shot (102305-6-0-0.wav). Each was mixed with white, pink, traffic (STRAFFIC), and crowd (PCAFETER) noise at 20, 10, and 0 dB, producing 4 clips × 4 noises × 3 SNRs = 48 analyses. The measured achieved SNRs agreed with their targets to numerical precision. The selected clips are illustrative rather than a random or representative dataset-wide sample.

For each mixture, the program computed the relative Frobenius distance ||F(noisy) − F(original)||_F / ||F(original)||_F separately for the log-Mel and MFCC time–frequency representations. These distances describe within-clip feature perturbation; they are not classification accuracies, speech-quality measurements, or causal measures of robustness. Table 4 summarizes the MFCC values averaged over the four selected clips. All 16 clip–noise combinations exhibited greater distance at lower SNR. At 0 dB, mean MFCC distance was highest under white noise (0.662), followed by pink (0.542), crowd (0.410), and traffic (0.330).

Table 4. Mean relative MFCC distance across the four selected clips (descriptive only).

Added noise	**20 dB**	**10 dB**	**0 dB**
White	0.335	0.485	0.662
Pink	0.236	0.371	0.542
Traffic	0.093	0.150	0.330
Crowd	0.104	0.250	0.410

The power-spectral-density curves in Figure 6 illustrate why equal total added-noise SNR does not imply identical spectral interference. White noise is comparatively flat, pink noise emphasizes lower frequencies, and the two recorded environments have irregular, scene-specific peaks and spectral slopes. Their plotted absolute levels should not be treated as a power-matched comparison; the noise mixer performs condition-specific RMS scaling before addition.

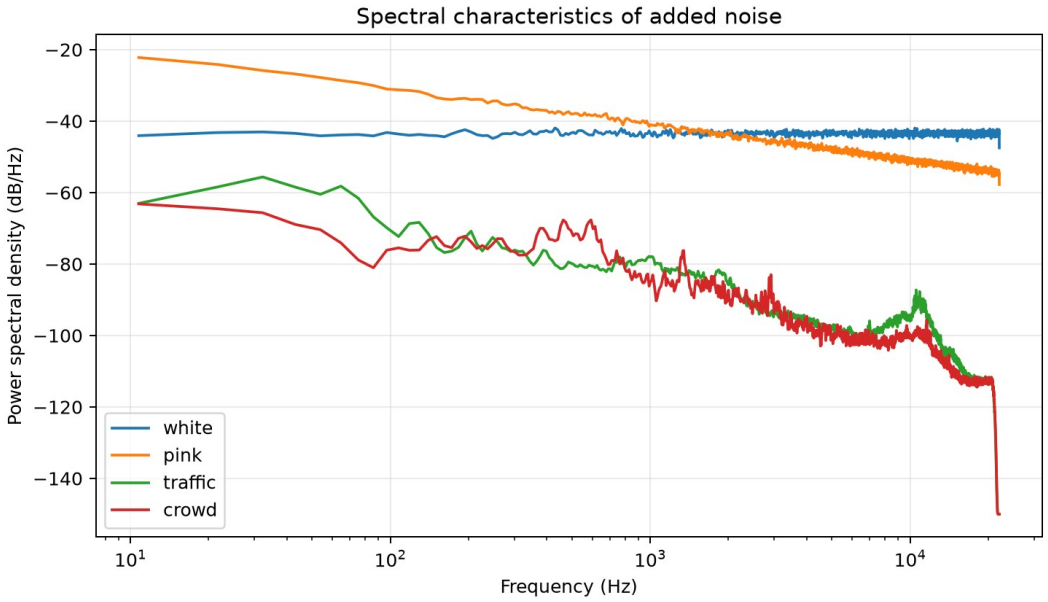


Figure 6. Power spectral densities of the four candidate noise sources, generated by the project signal-analysis program for the selected siren example. The curves show spectral character, not equalized mixture loudness.

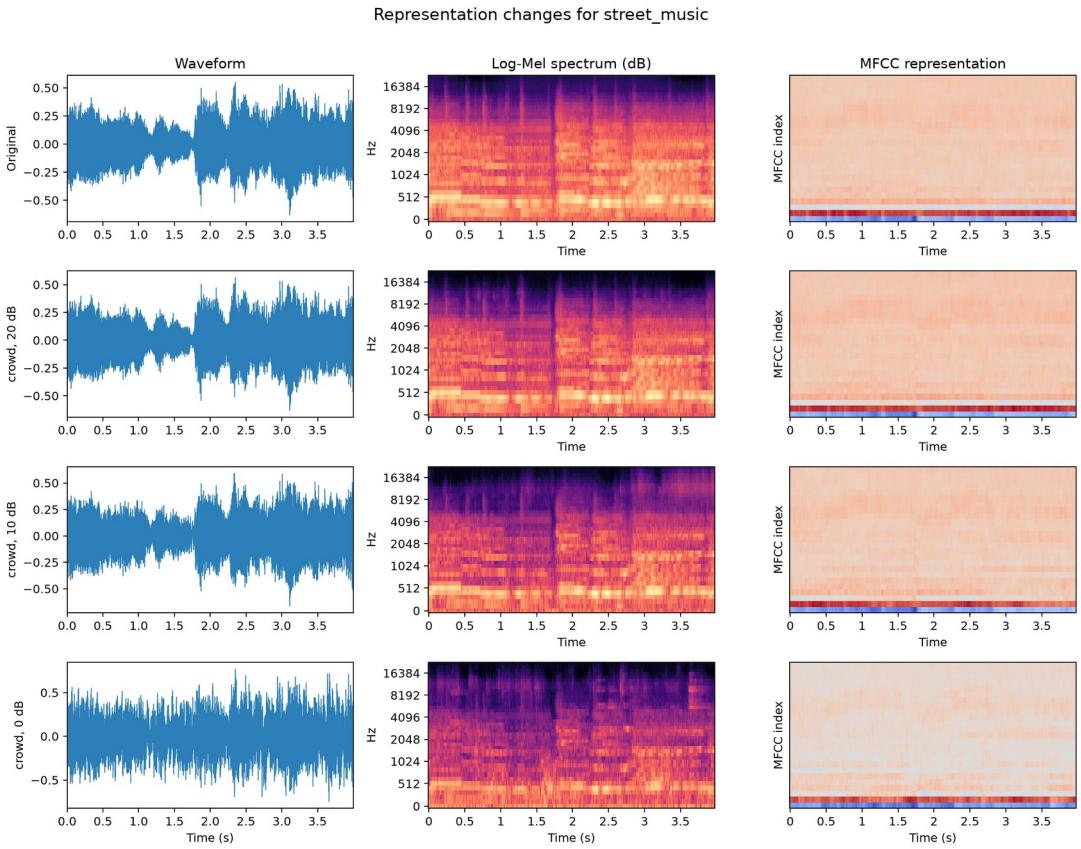


Figure 7. Full program-generated waveform, log-Mel and MFCC panels for the selected street_music recording (top row) and the same clip mixed with held-out crowd noise at 20, 10 and 0 dB (subsequent rows). This is one illustrative example, not a dataset-wide average.

The waveform and representations in Figure 7 visibly change under severe crowd interference, but the target sound also had ambient content before added noise. For engine_idling, the selected 0 dB crowd example showed a smaller MFCC distance (0.273) than the selected street_music example (0.454), while the full ten-fold classifier still performed poorly on engine_idling under crowd noise. A smaller feature-space shift alone therefore does not guarantee successful class recognition. The four-example supplement motivates, but cannot statistically establish, the mechanism of the observed class bias.

5. Discussion

5.1 What improves, and why the advantage varies

The first hypothesis is supported within the conditions tested: matching training and test noise is advantageous at severe SNRs. Relative to Original, White training increases accuracy on 0 dB white noise by about **27.97 pp**, while Traffic training increases accuracy on 0 dB traffic noise by about **13.27 pp**. In both instances, augmentation exposes the SVM to perturbations that resemble its evaluation condition. The increase is consistent with a decision boundary less dependent on particular noise-sensitive MFCC coordinates. However, this is an interpretation of the observed pattern rather than a direct measurement of the learned boundary.

The second hypothesis is also supported on the *equally weighted full suite*: mixed augmentation provides the best average accuracy and macro F1. Its broader set of synthetic and recorded training noises appears to trade some condition-specific specialization for more reliable performance across diverse tests. The marked degradation of Original at 0 dB further confirms that the baseline feature-classifier combination is sensitive to additive noise. This accords with earlier work emphasizing acoustic representation robustness in mismatched conditions [3] and augmentation-dependent performance differences [4].

The third hypothesis is not confirmed by the held-out PCAFETER experiment. Mixed did not beat Traffic on this specific unseen recording, and White augmentation was even below Original at 0 dB. Multiple explanations are plausible: the spectral or temporal character of the chosen traffic excerpts may happen to resemble components of the selected cafeteria noise; the random single-copy mixture may under-sample useful exposure to any one perturbation; or augmentation may shift decision boundaries unevenly across classes. The available metrics do not establish which explanation is correct. The supplementary spectrum plot confirms that these noise sources have different spectral distributions, but does not prove that spectral similarity explains the Traffic-versus-Mixed result; matched-power spectral comparisons, more held-out scenes, and a training-seed ablation would be needed to distinguish the alternatives.

5.2 Why macro metrics and per-class results matter

The class-level results point to significant brittleness hidden by a single average. Under strong held-out noise, the Mixed model over-predicts street_music and almost fails to identify some continuous machine-like sounds. This could arise when interference masks a target's distinctive time-frequency features, making its aggregate cepstral statistics resemble more common or broad-band classes. Salamon and colleagues reported related challenges from timbre similarity and real background interference [1]. The four supplementary waveform/log-Mel/MFCC examples demonstrate sizable representation changes as SNR falls, particularly for white and pink noise. However, they do not directly reveal the trained decision boundary, and the selected engine_idling example illustrates that smaller MFCC distance need not correspond to higher classification recall. The mechanism behind the class bias thus remains an informed hypothesis, not a validated causal explanation. Reporting macro F1 alongside accuracy and selected confusion matrices is therefore necessary rather than optional.

5.3 Methodological limitations and validity

Unequal training counts: Each augmented strategy includes an extra noisy copy of every training clip, unlike Original. Gains may reflect the combined effects of augmentation, increased training count, and changed class weighting in the SVM objective. An equal-size duplicate-original or resampling control would improve causal interpretation. Importantly, this limitation does not invalidate the *descriptive* differences measured, but does limit the attribution of those differences.

Four-sample acoustic illustration: The representation study uses one handpicked-by-rule Fold 1 clip per class, not random clips across folds. Its 48 entries reuse those four waveforms; they are not 48 independent audio observations and should not be used for dataset-wide significance tests.

Only one held-out scene and channel: PCAFETER is a genuinely excluded environment in the configured augmentation policy, yet it is only one selected microphone channel from one real recording. Its result should not be generalized to all cafes, crowds, or unseen acoustics without additional environments.

Single deterministic augmentation schedule: One seeded realization of the training noise type, segment, and SNR per clip does not quantify sensitivity to other random seeds, alternative segments, or repeat training runs. The fold-to-fold variation primarily reflects the UrbanSound8K splits, not all possible augmentation choices.

Interpretation of SNR: UrbanSound8K contains environmental background in the source waveform. Consequently, nominal mixture SNR uses *total original waveform* power as signal power. It does not quantify the perceptual salience or SNR of the ground-truth foreground event.

Approximate prior-method reproduction: The MFCC implementation and model-selection details use Python libraries and a documented 225-feature interpretation, rather than replicating every Essentia/Weka setting in [1]. Absolute accuracy differences relative to that paper should be considered implementation-sensitive, not an attempted leaderboard claim.

No full statistical independence of folds: The paired intervals use ten folds as the unit and appropriately avoid inflating sample size with repeated SNR conditions. However, their training sets overlap and the same source collection is repeatedly analyzed. The intervals are exploratory descriptors of the folds used here, not definitive probabilities for future deployment distributions.

5.4 Directly actionable next experiments

Three low-cost extensions would most improve confidence. First, train an Original-Duplicate control with the same number of original-feature training rows as augmented models. Second, evaluate multiple additional DEMAND environments excluded from training and vary mixing seeds to estimate sensitivity to noise selection. Third, expand the completed four-clip representation study to a stratified sample across classes and official folds, test the relation between per-clip MFCC change and out-of-fold prediction errors, and examine whether the observed street_music bias remains stable across multiple recordings.

6. Conclusion

This project completed a reproducible, official ten-fold UrbanSound8K experiment comprising four training policies, 13 evaluation conditions per policy, 520 fold-level rows, and 454,064 predictions. In the evaluated added-noise suite, mixed augmentation improved mean accuracy by 9.68 percentage points and macro F1 by 14.24 points relative to Original, while leaving average original-recording accuracy approximately unchanged. Stronger training-test noise matching produced clear local advantages, and the traffic-augmented classifier consistently surpassed Mixed on one held-out cafeteria environment. Severe crowd interference also produced a pronounced class-prediction bias toward street_music and poor recognition of several mechanical classes. A supplementary four-clip/48-mixture signal analysis demonstrates increasing MFCC and log-Mel perturbation as added SNR falls and illustrates the distinct spectra of synthetic and recorded noises, without establishing a dataset-wide causal explanation for errors. Thus, the research question has a qualified answer: additive noise augmentation improves this MFCC–SVM system under many tested conditions, but the benefit is conditional on noise domain and class, and more diverse augmentation does not guarantee superior performance on an unseen real environment. Future work should address the training-size confound, evaluate several held-out noise scenes, and connect acoustic representation changes to the observed class-specific errors.

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Reproducibility and submission links

Code repository: https://github.com/tongtongchen-540/elec5305-project-540334953

Video demonstration: https://youtu.be/T1r7gkJSBcs

Supplementary analytical outputs: results/full_experiment/ (classification) and results/signal_analysis/ (48 actual audio perturbations, per-clip representation plots, selected sample manifest and numerical distances).